MIL-T-22448(wep)

15 October 1963

MIlitary **SPECIFICATION**

Transformers, VARIABLE, PRECISION

This specification **has been approved by the Bureau of Naval Weapons, Department of the Navy.**

1. *SCOPE*

1.1 *scope.* - This specification describes the requirements for the procurement of three types of precision variable transformer. The precision variable transformer consists of an autotransformer having equally spaced taps, and a one-turn variable resistor - tapped portions of which are switched sequentially between taps of the autotransformer to provide interpolatlon. The switching is synchronized with rotation of the variable resistor wiper arm so that the voltage of the wiper varies smoothly, and is proportional to angle of rotation of the wiper arm throughout the entire voltage range of the precision variable transformer.

1.2 Classification. - The precision variable transformer shall be of the following types:

Type I	**BUWEPS** Dug. 1884650-1,	Recision **variable transformer; Output impedence** 45 ohms, **inPut** impedance 22,0000-
Type II	**BUWEPS Dwg. 1884650-2,**	Precision variable transformer; output impedence 130 ohms, **input** impedance 5o,ooo ohms.
Type III	Buweps dwg. **1884650-3,**	Recision variable transformer; output impedance 470 ohms, inPut impedance 103,000 ohms

2. APPLICABLE DOCUMENTS

FSC 1290

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2.1 The following specifications, standards, drawings, and publications, or such portion thereof as may be designated herein, of the issue in effect on the date of invitation of bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

ppp-b-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-corner
PPP-B-636	Boxes, Fiber

MILITARY

MIL-P-116	Preservation, mEthods Of
MIL-S-901	ShockProof Equipment Class HI (High Impact) Shipboard Application, Tests for
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Prep - arat ion for Delivery of
mIL-F-18870	Fire Central, Optical and Associated Equipment intended for shipboard Use, the General Specification for design and Procurement of
MIL-W-21927	weapons, Handling and Preparation for Delivery of, General Require - ments for

STANDARDS

MILITARY

MIL-STD-106	Mathentical Symbols
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechancal Vibrations of Shipboard Equipment

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DRAWINGS

NAVY DEPARTMENT

BUWEPS Dwg. 1834560-1	Transformer, Audio, Frequency Variable Linear Precision
BUWEPS Dwg. 1884650-2	Transfformer, Audio, Frequency Variable Linear Precision
BUWEPS Dwg. 1884655-3	Transformer, Audio, Frequency Variable Linear Precision

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

NAVWEPS Os 6341	Miscellaneous General Ordnance Deign Requirements

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuriong activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect On date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATION

Handboak H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government printing Office, Washinton 25, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE PUBLICATION

Consolidated Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Comnittee, 202 Union Station, Chicago 6, Ill.)

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AMERICAN TRUCking **ASSOCIATION, NC. PUBLICATION**

Motor Freight Classification Rules

(Application for copies should be addressed to the Issuing **Officer, American** Trucking **Association, Lat., l424** 16th Street, N.W., WashintOn 6, D. C.)

.3. REQUIREMENTS

3.1 **General** requirements.

3.1.1 Re-production samples. - Unless otherwise specified in the contract or order, pre-production samples of the precision **variable transformer shall be manufactured by the methods and procedures proposed for the production lot. These samples shall be tested as specified in section 4 herein and are for the purpose of determining that** the production items meet the requirements of this specification and the design. See 4.1.1.

3.1.2 Materials. - Materials restricted by **Specification** MIL-F-188770 shall not be used without specific approval of the procuring activity.

3.1.3 Construction, and dimensions. - Construction and dimensions for the precision variable transformer shall be in accordance with BUWEPS Dwg. 1884650.

3.1.4 Marking. - Markings of assemblies shall be in accordance with BUWEPS Dwg. 1884650, and the following requirement.

3.1.4.l Indexing. - The precision variable resistor shall be marked to indicate the percent output (ratio of output voltage to input voltage with 100 percent of the rated voltage applied across the two input terminals) corresponding to 50 percent rotation (midpoint of theoretical electrical angle).

3.1.5 Gages. - The **contractor shall provide himself with whatever gages are** necessary and adequate to **ensure that the material to which this specification applies will meet the requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages.** However, such procedure shall not relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the **satisfactory** fulfillment of the contract requirements, but Is intended to facilitate acceptance of all **components** and **assemblies by Naval** Find **Inspection Gages.**

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3.1.6 Threads. - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings, and the NOT-GO gaging practice shall be as set forth in the aforementioned Handbook H28.

3.1.7 Requests for deviation. - Each request that a deviation be authorized shall have been submitted to the Procuring Activity, and the deviation shall have been authorized by the Bureau Of NavaL Weapons or by a Naval Activity delegated such authority, prior to the production of the affected item, or prior to the performance of a particular operation on the item. The deviation shall apply only to this specific item or operation. Tne request shall define the suggested deviation In sufficient detail to permit the Procuring Actitity to evaluate its effect on Performance and reliability and the reasons for the suggested deviation shall be stated. Each request that a deviation be authorized, and the authorization, shall state specifically the extent of the departure, the identifying number of the item, the contractor (or bidder) and the contract number. A deviation is essentially temporary in character, and usually represents a departure limited to a specific period, to a specific number of items, or to the contract to which it applies. It may require an equitable reduction in contract price if a decrease in the cost of performance results therefrom.

3.1.8 Conflicting requirements. - conflicting requirements arising between this specification or any specification, publications, or drawings listed herein, shall be referred in Writing to the procuring agency or appointed agent for Interpretation, clarification and resolution (or correction).

3.2 Definitions. - Definitions applicable to this specification and associated drawings are as follOWS:

3.2.1 Shaft end play. - Shaft end play is the difference between the high and low dial indicator readings of the total axial travel of the shaft.

3.2.2 Shaft radial play. - Shaft radiaL play is the difference between the high and low dial indicator readings of the total radial shaft travel (sum of both directions).

3.2.3 Shaft run-out. - Shaft run-out is the difference been the high and low dial indicator readings of the shaft radial Movement at anY point on its exposed length when rotated with the housing held stationary.

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3.2.4 Concentricity. - Concentricity is the difference between the high and low dial indicator readings of mounting diameter radial movement with the shaft held stationary and the housing rotated.

3.2.5 Perpendicularity. - Perpendicularity is the difference between the high and low dial indicator readings of mounting diameter axial move - ment with the shaft held stationary and the housing rotated.

3.2.6 Starting torque. - Starting torque is the maximum torque re - quired to start rotation of the shaft.

3.2.7 Mechanical angle. - Mechanical angle is the total angle through which the shalt can be rotated between the mechanical stops.

3.2.8 Continuity angle. - Continuity angle is the angle through which the Shaft can be rotated while electrical continuity is maintained between the wiper and each end terminal of the precision variable transformer.

3.2.9 Theoretical electrical angle. - Theoretical electrical angle is an increment of shaft angle, specified on BUWEPS Dug. 1884650, through- out which it is required that the terminal conformity requirements be met.

3.2.10 Mechanical overtravel. - Mechanical overtravel is the angle of shaft rotation between either limit of theoretical electrical angle and the near mechanical stop .

3.2.11 *Terminal conformity.* - Terminal conformity is the maximum de- viation of actual output voltage from theoretical output voltage based on a straight line drawn through the points zero output voltage at the lower limit of theoretical electrical angle and output voltage equal to input vOltage at the upper limit of theoretical electrical angle. Terminal con- formity is expressed as a percentage in volts deviation Ion per volt of input voltage .

3.2.12 Electrical resolution. - Electrical resOlutiOn is a measure of the accuracy to which the precision varible *transformer output voltage may be* set with reference to the input voltage. The Shaft is moved through the smallest angular displacement which will produce a change in output voltage. The maximum incremental change in output voltage thus measured, observed anywhere in the continuity angle, divided by the input voltage and multiplied by 100, is the percentage electrical resolution.

3.2.13 Rated voltage. - Rated voltage is the maximum operating volt - age specified on BUWEPS dwg. 1884650.

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3.2.14 Tropical environment. - Tropical enviornment Is defined as conditIons of high humidity, high temperatures, and spore exposure, which present an opportunity for fungi growth.

3.3 Performance requirements and product characteristics. - The precision variable transfermer shall meet the following performance requirements and product characteristics:

3.3.1 **Mechanical requirements.**

3.3.1.1 shaft **end play. - When measured in accordance with 4.4.4.1, the shaft** end plaY **shall meet the requirements of BUWEPS Dwg.** 1884650. See 4.4.4.1.

3.3.1.2 Shaft **radial play. - When measured in accordance with** 4.4.4.2, **the shaft radial play shall meet the requirements of BUWePS dwg. 1884650. *See*** 4.4.4.2.

3.3.1.3 shaft run-out. - When measured in accordance with 4.4.4.3, **the shaft run-out at any point on its exposed length shall meet the requirements of BUWEPS dwg.** 19884650. See 4.4.4.3.

3.3.1.4 Concentricity. - When measured in accordance with 4.4.4.4, the mounting diameter radial run-out shall meet the requirements of BUWEPS Dug. 1884650. See 4.4.4.4.

3.3.1.5 Perpendicularity. - When measured in accordance with 4.4.4.5, the mounting face axial run-out shall meet the requirements of BUWEPS Dug. 1884650. See 4.4.4.5.

3.3.1.6 startIng torque .- When measured in accordance with 4.4.4.6,, the Starting torque shall meet the **requirements of BUWEPS Dug.** 1884650. see 4.4.4.6.

3.3.1.7 Mechanical angle. - **When measured In accordance with** 4.4.4.7, the **mechanical angle shall meet the requirements of BUWEPS Dug. 1884650. *See* 4..4.4.7.**

3.3.1.8 continuity ANGLE. - ***When the*** eclslon variable transformer IS tested in accordance with 4.4.4.8, the continuity angle shall be equal to the mechanical angle (specified in 3.3.1.7), and there shall be no abrupt changes in the resistance measured between the wiper and either end terminal as the shaft is rotated- See **4.4.4.8.** c

3.3.1.9 **Mechanical** overtravel. - When measured In accordance with 4.4.4.9, the Mechanical overtravel at **each limit of theoretical electrical angle shall meet the requirements of BUWEPS Dwg. 1884650. See 4.4.4.9.**

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3.3.1.10 Stop torque. - When the precision variable transformer is tested in accordance with 4.4.4.10, fallowing application to each mechanical stop **of shaft torque equal to the torque specified an BUWEPS Dwg. 1884650 for the stop torque test, the following conditions shall be satisfied:**

(a) The mechanical angle and the change in mechanical angle from the value measured prior to the stop torque test shall meet the requirements specified on BUWEPS Dug. **1884650 for** the stop torque test,

(b) The ***starting*** **torque shall meet the** requirements of BUWEPS Dug. 1884650,

(c) Continuity shall exist throughout the entire mechanical angle.

See 4.4.4.10.

3.3.2 Electrical requirements.

3.3.2.1 Dielectric strength and **insulation resistance.**

3.3.2.1.1 Dilectric strength. - When checked in accordance with 4.4.4.11.1, the dielectric strength shall meet the requirements of BUWEPS Dug. 1884.650. See 4.4.4.11.1.

3.3.2.1.2 Insulation resistance. - When measured in accordance with 4.4.4.11.2, the **insulation resistance shall meet the requiremmts of BUWEPS Dwg.** 1884650. See 4.4.4.11.2.

3.3.2.2 Terminal conformity. - **When measured in accordance with 4.4.4.12, the terminal conformity shall meet the requirements of BUWEPS Dug. 1884650.** See 4.4.4.12.

3.3.2.3 Center **tap. - When measured in accordance with 4.4.4.13, the voltage between the center tap and either end terminal shall meet the requirements of BUWEPS DWG.** 1884650. See 4.4.4.13.

3.3.2.4 Electrical resolution. -When measured in accordance with 4.k.4.l4, the electrical resolution shall meet the **requirements Of BUWEPS Dug.** 1884650. See 4.4.4.14.

3.3.2.5 Phase shift. - When measured in accordance with 4.4.4.15, the phase shift at 400 cycles per second, expressed in millivolts quadra**ture output per input volt, shall** meet the requirements of BUWEPS Dug. **1884650. See** *4.4.4.15.*

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3.3.2.6 Noise. - When measured in accordance with 4.4.4.16, the electrical noise shall meet the requirements of BUWEPS dwg. 1884650. see 4.4.4.16.

3.3.2.7 Input impedence. - When measured in accordance with 4.4.4.17, the input impedance shall meet the requirements of BUWEPS Dwg. 1884650. See 4.4.4.17.

3.3.2.8 Output impedance. - When mEASURED in accordance with 4.4.4.18, the output impedance shall meet the requirements of BUWEPS Dwg. 1884650. see 4.4.4.18.

3.4 Environmental requirements. - The precision variable transformer shall meet the other applicable requirements of this specification during or after tests defined or specified in 4.5.

3.4.1 Vibration

3.4.1.1 Vibration, transportation, storage, and hanfling. - The precision variable transform? shall withstand the tests specified in 4.5.1.1, and thereafter shall show no damage and shall meet all the requirements of section 3.

3.4.1.2 Vibration, operating condition. - The precision variable transformer shall withstand the type I vibration requirements specified in Standard MIL-STD-167, shall function during and after vibration tests, and thereafter shall show no damage and shall meet all the performante requirements and product characteristics of this specification. See 4.5.1.2.

3.4.2 Shock

3.4.2.l Shock, transportation, storage, and handling. - The precision variable tranformer shall withstand the tests specified in 4.5.2.1, and thereafter shall show no damage and shall meet all the requirements of section 3.

3.4.2.2 Shock, non-operating condition. - The precision variable transformer shall withstand the type A shock requirements specified in Specification MIL-S-901, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification. See 4.5.2.2.

3.4.3 Temperature, operating condition. - The precision variable transform shall withstand operational ambient temperatures of between minus 4 degrees Fahrenheit (minus 20 degrees centigrade) and 185 degrees

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3.3.2.6 Noise. - When measured in accordance with 4.4.4.16, the electrical noise shall meet the requirements of BUWEPS dwg. 1884650. see 4.4.4.16.

3.3.2.7 Input impedence. - When measured In accordance with 4.4.4.17, the input impedance shall meet the requirements of BUWEPS Dwg. 1884650. See 4.4.4.17.

3.3.2.8 Output impedance. - When mEAsured in accordance with 4.4.4.18, the output impedance shall meet the requirements of BUWEPS Dwg. 1884650. see 4.4.4.18.

3.4 Environmental requirements. - The precision variable transformer shall meet the other applicable requirements of this specification during or after tests defined or specified in 4.5.

3.4.1 Vibration

3.4.1.1 Vibration, transportation, storage, and handling. - The precision variable transform? shall withstand the tests specified in 4.5.1.1, and thereafter shall show no damage and shall meet all the requirements of section 3.

3.4.1.2 Vibration, operating condition. - The precision variable transformer shall withstand the type I vibration requirements specified in Standard MIL-STD-167, shall function during and after vibration tests, and thereafter shall show no damage and shall meet all the performante requirements and product characteristics of this specification. See 4.5.1.2.

3.4.2 Shock

3.4.2.1 Shock, transportation, storage, and handling. - The precision variable transformer shall withstand the tests specified in 4.5.2.1, and thereafter shall show no damage and shall meet all the requirements of section 3.

3.4.2.2 Shock, non-operating condition. - The precision variable transformer shall withstand the type A shock requirement specified in Specification MIL-S-901, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification. See 4.5.2.2.

3.4.3 Temperature, operating condition. - The precision variable transform shall withstand operational ambient temperatures of between minus 4 degrees Fahrenheit (minus 20 degrees centigrade) and 185 degrees

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Fahrenheit (85 degrees centigrade) in accordance with Specification MIL-F-18870, shall function during and after temperature tests, and thereafter shall show no damage or corrosion and shall meet all the performance requirements and product characteristics of this specification. See 4.5.3.1

3.4.4 Temperature, non-operating condition. - The precision variable transform shall withstand storage temperatures of **between minus** 80 degrees Fahrenheit (minus 62 degrees centigrade) and 167 degrees Fahrenheit (75 degrees centigrade) in accordance with Condition II of 0s 6341, and thereafter shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification. See 4.5.3.2.

3.4.5 Humidity. - The precision variable transformer shall withstand operational ambient humidities in accordance with Specification MIL-F-18870, and thereafter shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification. see 4.5.4.

3.4.6 Tropical enviornment. - The precision variable transformer shall withstand the tropical enviornmental conditions **specified in** Specification MIL-E-5272 under "Fungus Resistance Tests", and thereafter shall show no fungus growth, corrosion, or other damage, and shall meet all the performance requirements and product characteristics **of** this specification. See 4.5.5.

3.4.7 Operating life. - The precision variable transformer shall withstand the operating life test described in 4.5.6, and thereafter shall show not more than

(a) a 25 percent increase in starting torque,
(b) a 50 percent degradation in terminal conformity,
(c) a 25 percent increase in electrical noise,
(d) a 2 percent increase in output impedance,

and shall meet all the performance requirements and product characteristics of this specification with the exception that failure to comply with the requirements of 3.3.1.6, 3.3.2.2, 3.3.2.6, and 3.3.2.8 shall not be considered cause for rejection. see 4.5.6.

3.5 Clean-up. - PRior to and after final assembly, all parts, compoments and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as rosin flash, shall be removed.

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3.6 Workmanship. - The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, marking of parts and assemblies, rivettng, machine screw assembly, and freedom of' parts from burrs and sharp edges.

3.6.1 In addition to the examination and tests performed on the pre-production samples, the workmanship exhibited therein will be evaluated to determine acceptability. The approved standard of workmanship will thereby become a minimum production requirement for all units offered for acceptance.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or anY other inspection facilities and services acceptable to the Goveernment. The Government reserves the right to perfom anY of the inspection set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 pRe-production sample. - Unless otherwise specified for each contract or order, a representative sample of three precision variable transformers shall be manufactured by the methods and procedures proposed for the production lot. These shall be submitted for pre-production tests at an activity designated by the Bureau of Naval Weapons, for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting actiVity or the completion of tests on the pre-production samples accepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the Government.

4.1.2 Periodic production samples. - Unless otherwise specified by the procuring agency, the tests of 4.4 and 4.5 shall be evaluated as follows:

Step 1. - Upon initiation of production, select one (1) unit from each group of ten (10) units for submission to an activity designated by the procuring agency.

Step 2. - Upon satisfactory cOmpliance of each unit from three (3) consecutive groups, select one (1) unit from each group of twemty-five (25).

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step 3. – **Upon satisfactory compliance or each unit from** *three (3)* consecutive groups, **select one (1) Unit from each group Of fifty (50)** units. **Continue this procedure until a defective** unit is **found.**

step 4. - **When a defective unit is** found In any group, the group represented by the sample shall be rejected. Shift to step 1 and proceed as indicated. Disposition of groups represented by failng samples shall be made **by the Government.**

4.2 **Classification of tests. - The inspection and testing of the precision varable transformer shall be classified as follows:**

4.2.1 Acceptance tests. - These **tests shall** be accomplished on each precision varaible **transform submitted** *for* acceptance under contract. These tests are detailed In 4.4.

4.2.2 PRe-production and periodic production tests. - Pre-production and periodic production tests are those which are accomplished on samples selected as specified in 4.1.1 and 4.1.2 Which are representative of the production Of the precision variable *transformer after* the award of the contract, to determine that the **production meets the requirements of this specification. These tests are detailed in 4.5. Acceptance shall be based on no defects in sample. Failure of the sample to comply with these requirements will result in the reJection of the lot or the cessation of production* as determined by the** procuring **activity.**

4.2.3 Tests after delivery. - Unless otherwise specIfied in the contract , the Government reserves the right to conduct any or all the tests specified herein, as deemed necessary, after Installation of the precision variable transformer. Any defects discovered during or after these tests due to the contractor's failure to comply with drawings or specifications for this equipment shall be connected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

4.3 **Test equipment. - The following items of test equipment or equivalent are required to perform the tests** *set* forth in this specification.

4.3.1 End *play tester. - An end* **play tester or bench lathe capable** *of securing the* **precision variable transform in a vertical or horizontal POSItion and mounting a dial Indicator.**

4 3.2 Dial **indicator. -** *A* **dial indicator which can be mounted on the tester specified in 4.3.1,** and which measures with an accuracy of 0.0001 inch.

4.3.3 *Mechanical* loads.

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4.3.3.1 Reversing l/2-pound axial load. - A reversing l/2-pound load which can be applied axially to the shaft of the precision variable transformer.

4.3.3.2 Reversing l/2-pound radial load. - A reversing l/2-pound load which can be applied radially to the shaft of the precision variable transformer.

4.3.4 Torque wrenches.

4.3.4.1 2.0 ounce-inch torque wrench. - A torque wrench capable of applying and measuring a torque of 2.0 ounce-inches with an accuracy of plus or minus five percent.

4.3.4.2 35 pound-inch torque wrench. - A torque wrench capable of applying and measuring a torque of 35 pared-inches with an accuracy of plus or minus five percent.

4.3.5 Angle divider. - An angle divider having continuous 360-degree rotation with a maximum angular error of plus or minus *20* seconds of arc . The angular error shall include all positioning, coupling, and reading errors.

4.3.6 Ohmmeter. - A multi-range ohmmeter having mid-scale deflections of approximately 15, 1500, and 150,000 ohms, and a mid-scale accuracy of plus or minus 5 percent. The maxium current the OhmmEter applies on any range shall not exceed *10* milliamperes.

4.3.7 Whetstone bridge. - A drirect-current resistance bridge capable of measuring resistance from 10 to 1000 ohms with an accuracy of plus or minus *0.5* percent on all ranges. The maximum current the bridge applies on any range shall not exceed 10 milliamperes.

4.3.8 High potential tester. - A high potential test set capable of applying a minimum of 1000 volts at 60 cycles per second in 25 volt graduations which will cut out when load current exceeds 3 milliamperes. The rate of change output voltage shall not exceed 75 volts per second when the tester is turned on or off.

4.3.9 insulation resistance tester. - An insulation resistance test set capable of applying 530 volts direct current and measuring UP to 200 megohms.

4.3.10 Decade voltage divider. - An adjustable ratio transformer having a frequency range of 50 to 800 cycles per second, input voltage limit not less than 150 volts, calibrated to an accuracy of output-to-input ratio of plus or minus 0.001 percent, and adjustable in steps of 0.001 percent.

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4.3.11 oscilloscope. - A calibrated, direct-current coupled oscilloscope having a minimum input impedance of 250,000 ohms, a minimum response of plus or minus 5 decibels from direct current to 2,000 cycles per second, and a minimum sensitivity of 10 millivolts per inch.

4.3.12 Direct-current power supply. - A power supply capable of supplying a short-circuit current of 1 milliampere plus or minus 5 percent, and hating an open circuit voltage of 50 volts plus or minus 5 percent.

4.3.13 Alternating-current power supply. - A power supply capable of supplying a single-phase source of 400 cycle-per-second voltage adjustable from zero to 133 percent of precision variable transformer maximum input voltage per BUWEPS dwg. 1884650, having a frequency accuracy of plus or minus 5 percent, a total harmonic content not greater than 5 percent, and a volt-ampere rating of 10 volt-amperes minimum.

4.3.14 Voltmeter. - A multi-range alternating-current voltmeter, **0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 1, 10, 100,** and **300** volts full scale, **having an input impedance Of 100,00O ohms minimum,** and an accuracy of plus or minus 3 percent of full scale.

4.3.15 Milliammeter. - A direct-current mill-tar, 1 milliampere full scale, having internal resistance not greater than 50 ohms, and an accuracy of plus or minus 2 percent of full scale.

4.3.16 Resistor. - A precision non-inductive resistor, 100 ohms plus or minus one percent, one watt.

4.3.17 400-cycle filter. - A filter whose transmission, relative to that at 400 cycles per second, is down at least 40 decibels for frequencies of 200 cycles per second or lower, and for frequencies of 800 cycles per second or higher. The input impedance of the filter at 400 cycles per second shall not be less than 10,000 ohms.

4.3.18 2000-cycle filter. - A low-pass filter which attenuates frequencies above 2,000 cycles per second by at least 40 decibels.

4.3.19 Isolation transformer. - **A** shielded, audio-frequency isolation transformer having a turns ratio not greater than k to 1.

4,4 Acceptance tests.

4.4.1 The contractor shall conduct acceptance tests as specified in 4.2.1 to assure that the precision variable transformer is in compliance with the requirements of 3.1, 3.3., 3.5, and 3.6. Manufacturer's certification of materials specified in the applicable drawings shall be used in lieu of material tests.

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4.4.2 Test conditions. - Unless otherwise specified, the precision variable transformer shall be subjected to acceptance tests under the following conditions:

4.4.2.1 Temperature: Room ambient 65 to 95 degrees Fahrenheit (18 to 35 degrees centigrade).

4.4.2.2 Altitude: *Normal* ground atmospheric pressure.

4.4.2.3 Vibration: None.

"4.4.2.4 Humidity: Room ambient to 95 percent relative maximum.

4.4.3 Test and inspection equipment and facilities. - The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to **permit** performance of the required acceptance tests~

4.4.4 Test procedure. - The detailed test procedure listed in the following paragraphs shall be followed.

4.4.4.1 Shaft **end play. - With the precision variable transformer** mounted horizontally in the tester of 4.3.1 and the hoousing held stationary, the reversing l/2-pound load of 4.3.3.1 shall be applied axially to the shaft with the indicator of 4.3.2 mounted to measure axial movement of the shaft. The total indicator reading shall be observed to determine compliance-with the requirements of 3.3.1.1.

4.4.4.2 Shaft radial play. - With the precision variable transformer mounted horizontally in the tester of 4.3.1 and the housing held stationary, the reversing l/2-pound load of 4.3.3.2 shall be applied radially to the shaft between the flange and the mounting face with the indicator of 4.3.2 mounted to contact the shaft nor more the 1/16 inch from its end. The total indicator reading shall be observed to determine compliance with the requirements of 3.3.l.2.

4.4.4.3 Shaft run-out. - With the precision variable transformer mounted in the tester of 4.3.1 and the housing held stationary, the shaft shall be rotated with the indicator of 4.3.2 mounted to contact the shaft within 1/16 inch of its end. The total indicator reading shall be observed to determine compliance with the requirements of 3.3.1.3.

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4.4.4.4 Concentricity. - With the precision variable transformer mounted vertically in the tester of 4.3.1 and the shaft held stationary, the housing **shall** be rotated by hand with the indicator of 4.3.2 mounted to contact the mounting diameter of the unit. The total indicator reading **shall be observed** to determine compliance with the requirements of 3.3.1.4.

4.4.4.5 Perpendicularity. - With the precision variable transformer mounted vertically in the tester of 4.3.1 and the shaft held stationary, the housing shall be rotated by hand with the indicator of 4.3.2 mounted to contact the mounting face not more than 1/16 inch from its outside diameter. The total indicator reading shall be observed to determine compliance with the requirements of 3.3.1.5.

4.4.4.6 Starting torque. - With the torque wrench of 4.3.4.1 attached to the shaft of the precision variable transformer, torque shall be applied by hand to the housing and Increased slowly until it starts to rotate. This operation shall be repeated for several different wiper positions for both directions of rotation. The greatest value of torque required to start rotation of the housing shall be noted to determine **compliance** with the requirements of 3.3.1.6, and recorded.

4.4.4.7 Mechanical angle. - Using the angle divider of 4.3.5, and turning the shaft in one direction to avoid backlash, the mechanical angle from one stop to the other stop shall be measured to determine compliance with the requirements of 3.3.1.7, and recorded.

4.4.4.8 Continuity angle. - **Using** the angle divider of 4.3.5, and With the ohmmeter of 4.3.6 connected between the OUT terminal and one end terminal (either cw or ccw), the shaft **"shall be** rotated slowly over the entire mechanical angle while observing the ohmmeter reading, and the procedure shall then be repeated for the ohmmeter connected between the OUT terminal and the other end terminal to determine compliance with the requirements of 3.3.1.8.

4.4.4.9 Mechanical overtravel. - Using the circuit of figure 1, and with the shaft positioned by the angle divider of 4.3.5, the decade voltage divider shall be set to give the percentage output voltage marked on the precision variable transformer as corresponding to 50 percent rotation (midpoint of theoretical electrical angle), and the shaft shall be positioned to produce a null reading on the voltmeter. The shaft angles corresponding to the upper and lower limits of theoretical electrical angle shall be computed by respectively adding to and subtracting from the shaft angle corresponding to the null reading on the voltmeter an angle equal to one-half the theoretical electrical angle specified on BUWEPS Dwg. 1884650. The angle between each of the two limits of the theoretical electrical angle and the nearest mechanical stop shall be measured to determine compliance with the requirements of 3.3.1.9.

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4.4.4.10 Stop torque. - Using the torque wrench of 4.3.4.2, and with the wiper against either stop, the torque specified in 3.3.1.10 shall be applied. The wiper shall then be turned against the other stop, and the specified torque shall be applied. Following the torque applications, shaft mechanical angle, continuity throughout the entire shaft mechanical angle, and starting torque **shall** be checked to determine compliance with the requirements of 3.3.1.10.

4.4.4.11 Dielectric strength and insulation resistance.

4.4.4.11.1 Dielectric strength. - Using the high potential tester of 4.3.8, the dielectric test voltage specified on BUWEPS Dwg. 1884650 shall be applied between all terminals connected together and the shaft. The tester shall be observed for cut out to determine compliance with the requirements of 3.3.2.1.l.

4.4.4.11.2 Measurement of insulation resistance. - Immediately after application of' high potential voltage in accordance with 4.4.4.11.1, the insulation resistance tester of 4.3.9 shall be used to measure the resistance between all terminals tied together and the shaft to determine compliance with the requirements of 3.3.2.1.2. The 500 -volt potential of the insulation resistor tester shall be applied for 60 plus or minus 5 seconds.

4.4.4.12 Terminal conformity. - With the precision variable transformer connected as shown in figure 1, and with the upper and lower limits of the theoretical electrical angle determined as specified iii 4.4.4.9. the setting or the decade voltage divider which gives a null reading of the voltmeter shall be recorded every 90 degrees for types I and II, and every 36 degrees for precision variable resistor type III throughout the entire theoretical electrical angle, starting at either of its Limits. The quantity A J shall be computed for each shaft angle where

$$L\ J\ \frac{e_a\ \&\ e_L}{O_t} - n_J$$

θ_j = Angle of shaft measured from lower limit of theoretic- electrical angle (degrees),

o_L = Lower limit of theoretical electrical angle (degrees),

9_t = Theoretical electrical angle specified on BUWEPS Dwg. 1884650 (degrees),

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n_j = Setting of decade voltage divider (volts per volt).

The greatest A J shall be noted and multiplied by 100 to determine compliance with the requirements of 3.3.2.2, and recorded. Mathematical symbols shall be interpreted in accordance with Standard MIL-STD-106.

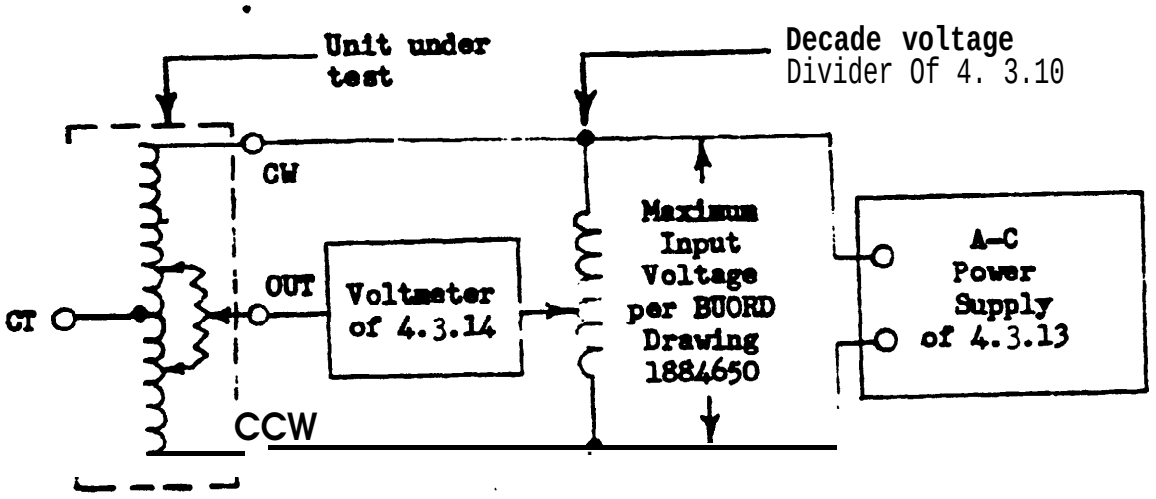
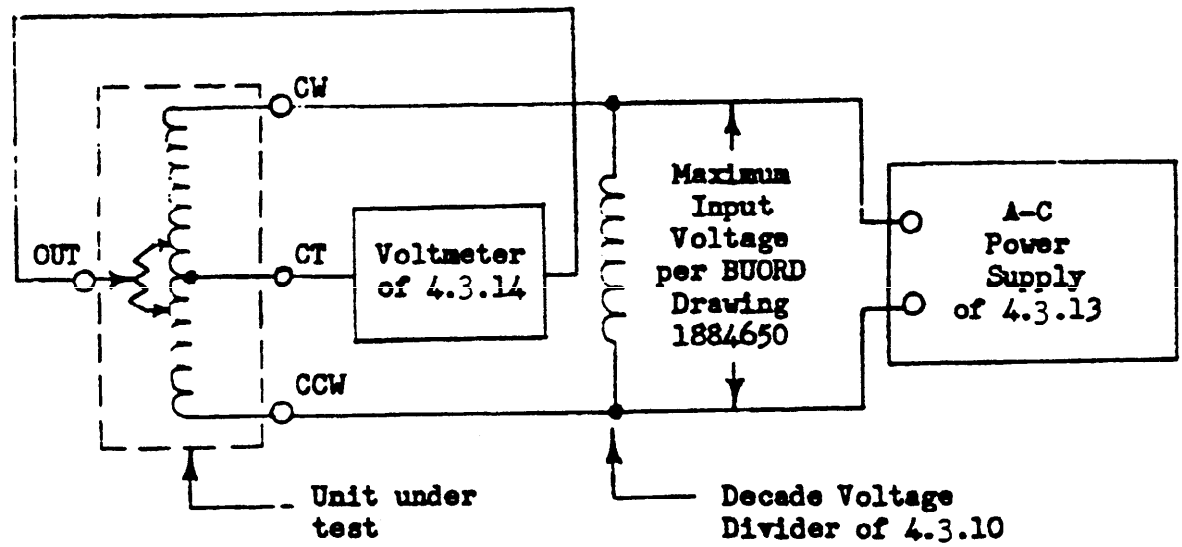


Figure 1. Terminal Conformity and Electrical **Rasolution** Test Connection

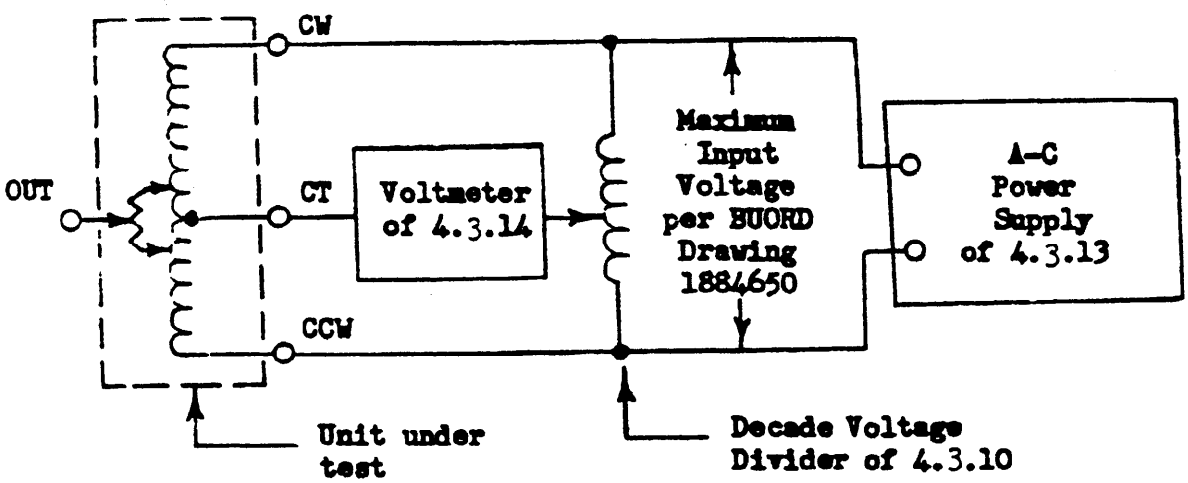
4.4.4.13 **Center tap. -** With the **precision variable transformer connected as shown in figure 2A, the** shaft shall be positioned to give a null reading **on the voltmeter.** Without disturbing **the setting of the** shaft, **the** voltmeter **connect Ions shall be changed to those** shown in figure 2B, and the decade volt- **age divider shall be 'adjusted to give** a null reading on the voltmeter. The **setting of the decade voltage divider** shall be subtracted from 0.500000 and **the difference shall be multiplied by** 100 to determine compliance with the **requirements of 3.3.2.3~**

4.4.4.14 Electrical resolution. - With the precision variable transformer connected as shown in figure 1, the shaft shall be rotated in one direction to take up the backlash. Without disturbing the shaft position, the decade voltage divider shall be adJusted to give a null reading on the voltmeter . The shaft shall then be rotated slowly in the same direction in which backlash was taken up through the smallest anguler displacement that produces a change in the voltmeter reading, and the change in voltmeter reading shall be noted. This procedure shall be repeated for several different shaft angular positions, and the maximum change in output voltage thus measured shall be divided by the input voltage and multiplied by 100 to determine compliance with the requirements of 3,3.2.4.

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A



B

Figure 2. Center-Tap Test Connections

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4.4.4.15 Phase shift. - With the precision variable transformer connected as shown in figure 3, and the wiper set to give a voltage between points G and A of 75 plus or minus 5 percent of the voltage between points G and C, the tap on the decade voltage divider shall be adjusted to minimize the voltage between points G and B. The voltage in millivolts between points G and B, referred to the primary of the isolation transformer, shall be divided by the voltage in volts between points G and A to determine compliance with the requirements of 3.3.2.5.

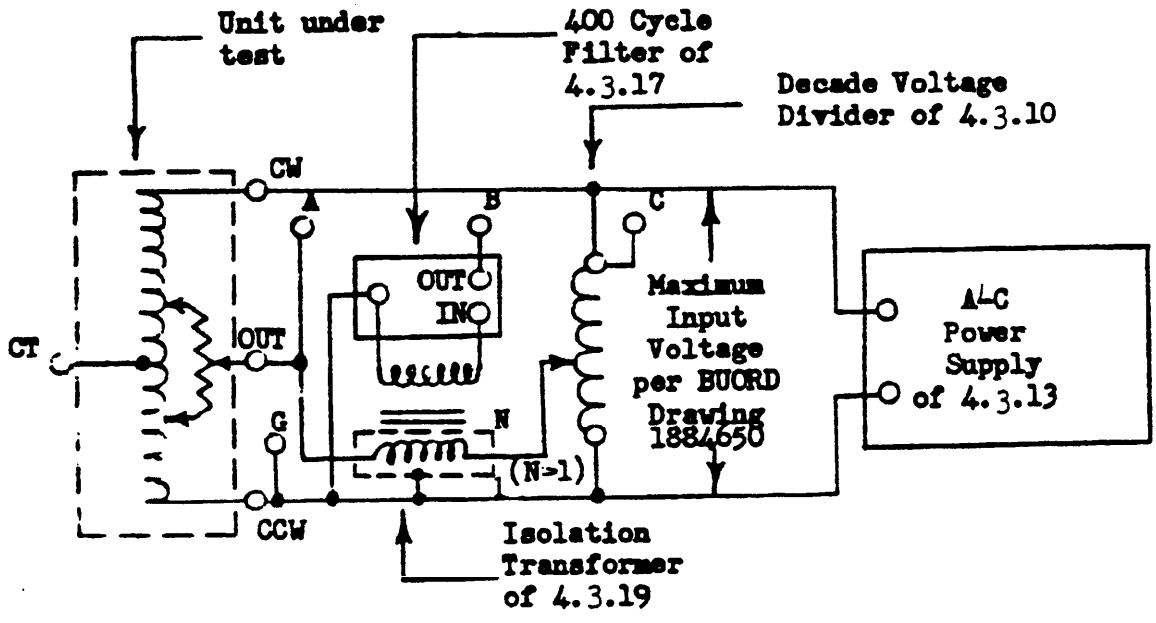


Figure 3. Phase Shift Test Connections

4.4.4.16 Noise. - With the precision variable transformer connected as shown in figure 4, the shaft shall be rotated in each direction at the speed specified on BUWEPS Dwg. 1884650 for the noise test, and the maximum peak-to-peak voltage shall be observed to determine compliance with the requirements of 3.3.2.6, and recorded.

4.4.4.17 Input impedance. - With the precision variable transformer connected as shown in figure 5, voltage E_R shall be observed as the shaft is rotated. This voltage should vary between an upper and a lower limit as the shaft is rotated. Input impedance shall be computed for each input test voltage (Ev) specified on BUWEPS Dwg. 1884650 to determine compliance with the requirements of 3.3.2.7 using the formula

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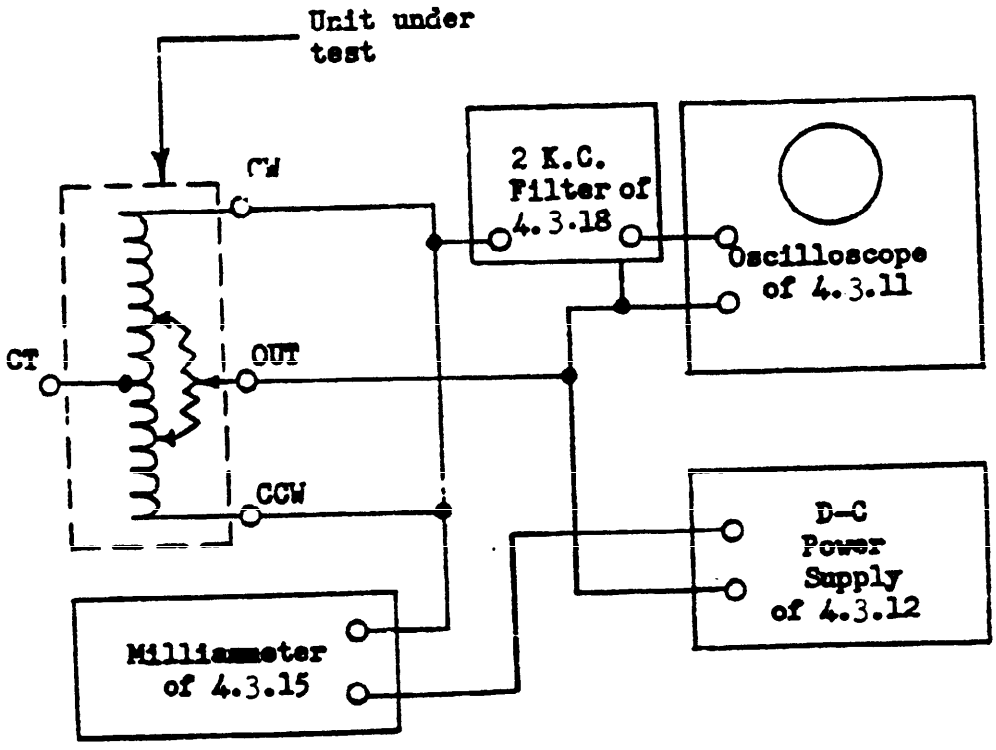


Figure 4. Noise Test Connections

$$Z_{in} = 100 \frac{E_v}{E_R}$$

where Z_{in} = input impedance (ohms)

E_v = lower limit of voltage between CW and CCW terminals of precision variable transformer (volts). See figure 5,

E_R = output voltage of filter 4.3.17 referred its input terminals (volts) when shaft position is that corresponding to E_v. See figure 5.

Mathematical symbols shall be interpreted in accordance with Standard MIL-STD-106.

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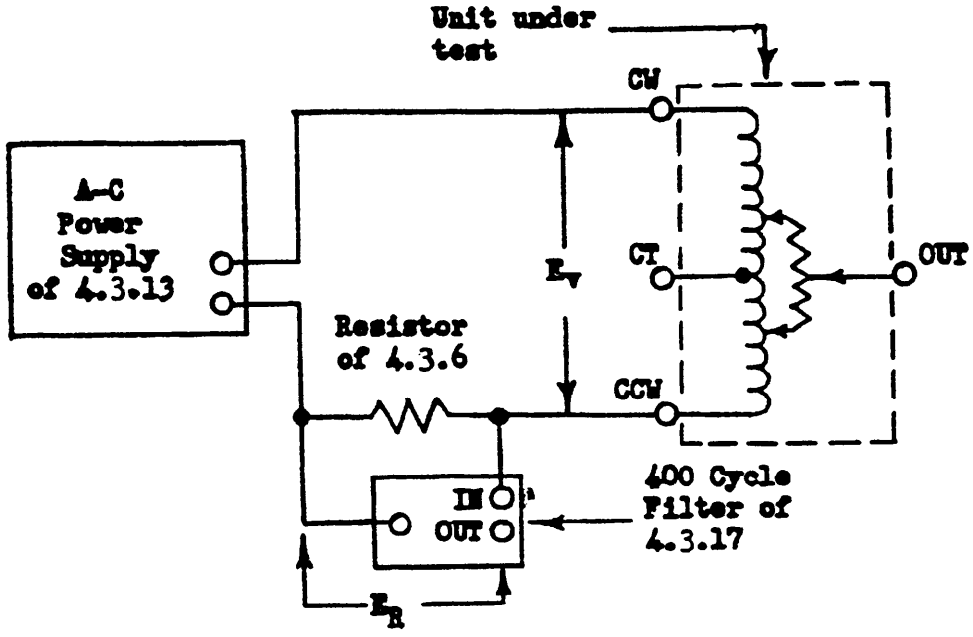


Figure 5. Input Impedance Test Connections

4.4.4.18 Output impedance. - With the precision variable transformer connected as shown in figure 6, the shaft shall be rotated to bring the wiper into contact with the CCW terminal. The resistance between the CCW and OUT terminals will be lower for this shaft position than for any other. The shaft shall be rotated clockwise from this position until a maximum resistance reading is obtained. The resistance between the CCW and OUT terminals corresponding to this shaft position shall be measured using the Wheatstone bridge of 4.3.7 to determine compliance with the requirements of 3.3.2.8.

4.4.4.19 Preservation, packing, and packaging. - Preservation, packing and packaging of the precision variable transformer shall conform to section 5 of this specification.

4.5 Pre-production and periodic production tests.

4.5.1 Vibration

4.5.1.1 Vibration, transportation. - The precision variable transformer, packaged and packed as specified in section 5., shall be vibration tested as specified in Specification MIL-W-21927.

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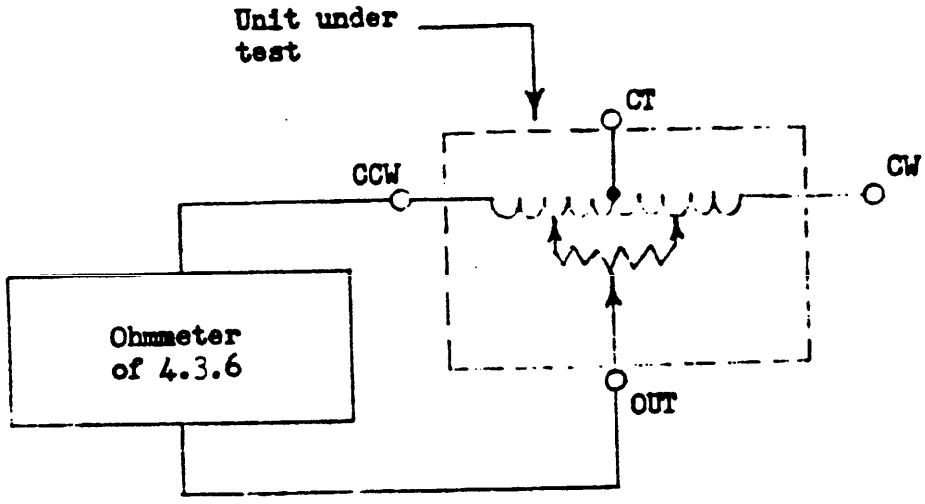


Figure 6. Output Impedance Test Connections

4.5.1.2 Vibration, operating condition. - The precision variable transformer, rigidly mounted in a suitable vibration fixture, shall be subjected to vibration tests so as to meet the requirements specified in 3.4.1, and shall meet all the performance requirements and product characteristics of this specification during and after the vibration tests.

4.5.2 Shock

4.5.2.1 Shock, transportation, storage and handling. - The precision variable transformer packaged and packed as specified in section 5, shall be subjected to the appropriate rough handling tests specified in Specification MIL-W-21927.

4.5.2.2 Shock, non-operating condition. - The precision variable transformer, rigidly mounted in a suitable shock fixture shall be subjected to shock tests so as to meet the requirements specified in 3.4.2.2and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.3 Temperature

4.5.3.1 Temperature, operating condition. - The precision variable transformer shall be subjected to temperature tests so as to meet the requirements specified in 3.4.3, and thereafter shall meet all the performance requirements and product characteristics of this specification.

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4.5.3.2 Temperature, non-operating condition. - The precision variable **transformer shall be** subjected to temperature tests so as to meet the requirements specified in 3.4.4, and thereafter shall meet all the performance requiremnts and product characteristics of this specification.

4.5.4 Humidity. - The precision **variable transformer shall be sUbJected to humidity tests so as to meet the requirements specified in 3.4.5, and thereafter shall meet all the performance requirements and product characteristics of this specification.**

4.5.5 **Tropical** environment. - The precision variable transformer shall be exposed to fungus tests so as to meet the requirements specified in 3.4.6, and thereafter shall meet all the **performance** requirements and product characteristics of this specification.

4.5.6 **Operating life. - The output impedance of** the precision variable transform shall be measured (see 4.4.4.18). This output impedance value shall be used as a reference for this test. The unit shall be mounted with standard hardware, leaving a minimum of 2 inches of free space around the unit. The shaft shall be oscillated at the rate of 240 degrees per second for a total of 500,000 oscillations. This oscillation shall encompass 360 degrees of shaft rotation, and the limits of shaft position during oscillation shall be 1567 plus or minus 4 degrees, and 1927 Plus Or minus 4 degrees from either limit of theoretical electrical angle. measurements of output impedance shall be made after 1000, 5000, 50,000, 100,000, 250,000 and 500,000 cycles to determine compliance with the output impedance requirements of 3.4.7. If the increase in output impedance is less than 2 percent at the end of 500,000 cycles, then the starting torque, terminal conformity, and electrical noise shall be measured (see 4.4.4.6, 4.4.4.12, and 4.4.4.16) to determine compliance with the starting torque, terminal. conformity, and electrical noise requiremmts of 3.4.?.

4.6 **Waiver of tests. - Tests specified in 4.5.1.1 and 4.5.2.1 may be waived by the procuring agency for** level C **packs** *only*, provided the con**tractor guarantees safe** delivery to the using activity.

4.7 Functional tests. - At the conclusion Of the **tests specified** in 4.5.1.1 and 4.5.2.1, fuctional other tests, as applicable, shall be conducted on the transformers to determine freedom from operational malfunction and compliance with all the requirements of section 3.

5. PREPARATION FOR DELIVERY

5.1 Preservation And packaging.

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5.1.1 level A. - In addition to the following, level A packaging shall be in accordance with MIL-p-17555.

5.1.1.1 Cleaning, drying and preservative application. - Not applicable.

5.1.1.2 Unit packaging. - Unit packaging of the transformers shall be in accordance with Method IIe of Specification MIL-F-116. The unit package shall contain one (1) transformer.

5.1.1.3 Intermediate package. - Not applicable.

5.1.1.4 Cushioning. - The unit package, intermediate package {when used) and pack, shall incorporate sufficient cushioning material, bracing, or other adequate shock **absorbing devices to ensure that-the transformers will meet all the** requirements of section 3 after being tested in accordance with 4.5.1.1 and **4.5.2.1.**

At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent. The pH range of the material shall be between 6.o and 8.o.

5.1.2 Level B. **- Not applicable.**

501.3 Level C.

5.1.3.1 Cleanlng, drying and preservative **application. - Not ap-**plicable.

5.1.3.2 Unit packaging. - Unit packaging of the transformers shall be in accordance with Method III of Specification MIL-P-116. The unit package shall contain one (1) transformer.

5.1.3.3 Intermediate package. - Not applicable.

5.2 Packing.

5.2.1 Level A. - In addition to the following, level A packing shall be in accordance with Specification MIL-P-17555.

5.2.1.1 Exterior containers. - Transformers packaged as described in 5.1.1.2 shall be **packed in a** wood box in accordance with Specification ppp-B-601, Overseas type or PPP-B-621, **(gross weight** not to exceed 150 Ibs) or in fiber boxes in accordance with Specification PPP B-636, class 2 (gross weight not to exceed 70 lbs).

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5.2.2 Level B.

502.2.1 Exterior containers. Transformers **packaged as described in 5.1.1.2 shall be packed in a wood box In accordance with Specification PPP-B-601,** *domestic* **type** *Or* PPP-B-621, class 1 (gross weight not to exceed 150 **lbs) or a fiber boxes In accordance with** Specification PPP-B-636, class 1 (gross weight not to **exceed 70 lbs).**

5.2.3 Level C. - Transformers packaged in accordance with 5.1.3.2 shall be packed afford protection against damage during shipment from the **supply source to the first receiving** actitity **for** Immediate use. COn-**tainers shall comply** with **Consolidated Freight Classification Rules or other common Carrier regulations** applicable **to the mode of transportation.**

5.3 Marking.

5.3.1 **special markings. - Hone, unless otherwise** specified.

5.3.2. Normal markings. . **In** addition to the markings required by contract or order, unit packages, intermediate packages (when used), and shipping **containers shall be marked in accordance with the** requirements of MIL-sTD-129, including serial numbers.

6. NOTES

6.1 Intended **use. - The precision -table** transformer is intended for use in the Angle SolverS **Mark 18 Mod 2 and Mod** 3, the Fire **Control Sys - tem Mark 112 Mod 1, the Sonar Receiver Sets M/BQG-l,** -2, **-3, -4,** or other **suitable application.**

6.2 Ordering data. - **Procurement documents should specify the specific title, number, and date Of this specification and exceptions to this specification; applicable drawings and other documents; selection of applicable levels of packing; and guarantee clause (See 6•5) .**

6.3 Criteria for use Of proper level Of preservation, packaging, and packing shall be as follows :

For level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For level B. This level shall be used only when it is **definitely known** that the item will be held in covered storage in overseas . locations for 6 months or less or in domestic locations for an indefinte period.

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For level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for *immediate use at the first receiving* activity.

6.4 The attention of the contracting officer is invited to the Quality assurance provisions and options in Specification MIL-p-116 and to Par. 4.6.

6.5 *Guarantee. - The* contract or order shall include a requirement for a guarantee stating that any precision variable transformer which fails to *meet the* requirements of this specification within one year after delivery because of a defect in material or workmanship or because of the vendor's failure to comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor at no cost to the Government.

6.6 The contract or order shall include a sufficient number of extra items to cover production and pre-production testing samples.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any waY supplied the said drawings, specification, or other data is not to be regarded by indication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented inventions that may in any way be related thereto.

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BUREAU OF NAVAL WEAPONS
WASHINGTON 25, D.C.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/9 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 49-R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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